



PFAM PRECISION
FLANGE & MACHINE

13123 West Hardy Rd
Houston, TX 77060
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Material Test Report

Material Test Report#
3770
Date
3/13/2025

Sold To		DNO W.L.P.																								
Order#	Line#	Quantity	PO#	Tag#	Batch																					
2319477	1	2	662674791	COST CENTER 107334	F25A5S																					
Heat Code	Specification	Revision	Year	Description																						
R24R9	ASA105N/ASA350 LF2 CL1	2023		3.13 2000 RTJ BLIND FLANGE A105N 8B																						
Chemical Analysis																										
Heat Code	C%	CP%	CU%	MN%	MO%	NI%	P%	SI%	S%	V%	AL%	NE%	SN%	N%	B%	CAV%	W%	TP%	CO%	AS%	SB%	HP%	PS%	ZR%	BI%	GE%
R24R9	0.18	0.10	0.15	1.17	0.03	0.06	0.007	0.29	0.004	0.034	0.025	0.003	0.005	0.0087	0.0001	0.0006	0.004	0.002	0.006	0.002	0	1.7PPM	0	0	0	0.42
Mechanical Properties																										
Heat Code	Yield PSI	Tensile PSI	Elong%	R of A	HB	GRAIN																				
R24R9	54700	80200	33.9	64.8	149	10																				
Impact Test																										
Heat Code	CHARPY V-NOTCH	CUNAV	CUN TEMP	LATERAL EXPANSION	SHEAR																					
R24R9	45 / 47 / 51	48	-50F	36 / 38 / 40	40 / 40 / 40																					
Heat Treatments																										
Heat Code	Heat Treatment																									
R24R9	NORMALIZED @ 1675°F FOR 2 HRS. COOLING STILL AIR TO 1000°F																									
Supplements																										
Heat Code	Supplement																									
R24R9	CONFORMS TO NACE MR0175/ISO 15156:2015 & MR0103/ISO 17946:2015. CHARPY V-NOTCH IMPACT RESULTS ARE STATED IN FT. LBS. MELTED & MANUFACTURED IN THE USA. FINE GRAIN PRACTICE. VACUUM DEGASSED. FULLY KILLED. LATERAL EXPANSION IS IN MILS. YIELD & TENSILE RESULTS OBTAINED FROM TEST COUPONS FROM THE SAME HEAT TREATMENT BATCH OF FITTINGS.																									

I HEREBY CERTIFY THAT THE INFORMATION SUPPLIED ABOVE IS TRUE AND CORRECT AND MATCHED THE INFORMATION KEPT IN OUR FILES. THE MATERIAL IS ALSO IN COMPLIANCE WITH ALL APPLICABLE SPECIFICATIONS. ANY ALTERATION IS PUNISHABLE UNDER FEDERAL LAW. DIN 50049 3.1.B EN 10204 3.1 ISO 16474 3.1.B ISO 9001:2015

Approved By
Sam Brown